AI-Powered CRM Platform

Abstract

The **AI-Powered CRM Platform** is an intelligent relationship management system designed to support business operations across different sectors. In the current implementation, the platform focuses on two sectors: **Agriculture and E-Commerce**. It provides multilingual support, AI-based chat and voice assistance, personalized recommendations, sentiment analysis, and next-best-action capabilities.

The platform uses **Retrieval-Augmented Generation (RAG)** with sector-specific knowledge bases and personalized user data to generate relevant and dynamic responses. It stores user interactions, conversations, orders, and activities securely, allowing the system to provide personalized support and recommendations. Administrators can manage products, users, orders, support requests, knowledge bases, and analytics through dedicated dashboards.

Functional Requirements

1. User Management & Security

- **FR1:** The system shall provide secure user registration, login, and logout.
- **FR2:** The system shall maintain user profiles and interaction history.
- **FR3:** The system shall securely protect and isolate user-specific data.

2. Language & Sector Selection

- **FR4:** The system shall support English, Telugu, and Hindi.
- **FR5:** Users shall be able to select their preferred language.
- **FR6:** Users shall be able to select the required sector: **Agriculture or E-Commerce**.
- **FR7:** The system shall provide localized UI and AI responses based on the selected language and sector.

3. Agriculture & E-Commerce

- **FR8:** The Agriculture sector shall provide products such as seeds, fertilizers, and farming instruments.
- **FR9:** The E-Commerce sector shall provide products such as laptops, mobiles, tablets, earphones, smartwatches, cameras, and accessories.
- **FR10:** Users shall be able to search, view, add products to cart, purchase products, and track orders.
- **FR11:** The system shall maintain order history and order status.

4. AI Assistant & Voice Agent

- **FR12:** The system shall provide AI-based text assistance for user queries.
- **FR13:** The system shall provide AI-based voice assistance.
- **FR14:** The voice agent shall perform speech-to-text, identify language and intent, retrieve relevant information, and generate responses.
- **FR15:** The system shall support text-to-speech for voice-based responses.

5. Personalized AI & RAG

- **FR16:** The AI shall use relevant profile data, search history, viewed products, wishlist, cart, orders, reviews, and previous conversations.
- **FR17:** The system shall maintain separate knowledge bases for Agriculture and E-Commerce.
- **FR18:** The system shall use RAG to retrieve relevant information from the appropriate knowledge base.
- **FR19:** The AI shall generate dynamic responses using personalized data, interaction history, and sector-specific knowledge.

6. Sentiment Analysis & Next Best Action

- **FR20:** The system shall analyze user interactions to identify sentiment and satisfaction levels.
- **FR21:** The system shall identify dissatisfaction, frustration, or positive interactions.
- **FR22:** The system shall determine an appropriate next-best action based on the user's situation.
- **FR23:** Possible actions shall include providing order information, recommending products, creating support requests, sending notifications, processing replacement/refund requests, or escalating to human support.

7. Conversation & Support Management

- **FR24:** The system shall securely store user conversations and interactions.
- **FR25:** The system shall maintain conversation history for future interactions.
- **FR26:** Users shall be able to raise support requests for their issues.
- **FR27:** Unresolved issues shall be escalated to human support when required.

8. Admin Management & Analytics

- **FR28:** The system shall provide administrative dashboards for Agriculture and E-Commerce.

- **FR29:** Administrators shall manage users, products, inventory, orders, and support requests.
- **FR30:** Administrators shall manage sector-specific knowledge bases and monitor conversations.
- **FR31:** The system shall provide analytics for users, orders, products, conversations, sentiment, and support activities.